

Tutorial 1

Prove the following statements using the definition of convexity.

- (a) If $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ are convex and $\gamma_i \geq 0$, then $F(x) = \sum_{i=1}^r \gamma_i f_i(x)$ is convex.
- (b) If $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is convex, $M \in \mathbb{R}^{m \times d}$, and $q \in \mathbb{R}^m$, then $F(x) = g(Mx + q)$ is convex.
- (c) If $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ are convex, then $F(x) = \max_{1 \leq i \leq r} f_i(x)$ is convex.

1. Determine whether each statement is *True* or *False*. For a true statement, give a short justification. For a false statement, give a counterexample and verify it.

- (a) If $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex, then $h(t) = g(1 - 2t)$ is convex.
- (b) If $f_1, f_2 : \mathbb{R} \rightarrow \mathbb{R}$ are convex, then $h(t) = \min\{f_1(t), f_2(t)\}$ is convex.
- (c) A symmetric matrix whose entries are all strictly positive is positive definite.
- (d) If $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice continuously differentiable and t^* is a strict local minimizer, then $f''(t^*) > 0$.
- (e) For differentiable $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $\nabla f(x) \neq 0$, $p = -3\nabla f(x)$ is a descent direction at x .
- (f) If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth with $L > 0$ and $\nabla f(x) \neq 0$, then

$$f\left(x - \frac{1}{L}\nabla f(x)\right) < f(x).$$

2. For a linear prediction model $a_i^T x$, let

$$a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, y_1 = 1, \quad a_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, y_2 = 2, \quad a_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, y_3 = 1.$$

We minimize the mean squared-error objective

$$F(x) = \frac{1}{2n} \sum_{i=1}^n (a_i^T x - y_i)^2, \quad n = 3.$$

- (a) Write F explicitly in terms of x_1, x_2 , and as $F(x) = \frac{1}{6} \|Ax - y\|^2$. State the dimensions of A and y .
- (b) Use the chain rule to find $\nabla F(x)$ in matrix form, then write its two components explicitly.
- (c) At $x^{(0)} = (0, 0)^T$, calculate the directional derivatives along

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Which is a descent direction? Find the steepest descent direction among all unit vectors.

3. Consider

$$f(x_1, x_2) = x_1^2 - 2x_1x_2 + \frac{2}{3}x_2^3.$$

- (a) Compute $\nabla f(x)$ and $\nabla^2 f(x)$, and find all stationary points.
- (b) Classify each stationary point as a local minimizer, a local maximizer, or neither. You may test a Hessian using $z^T \nabla^2 f(x) z$; no eigenvalue calculation is required.
- (c) Does f have a global minimizer on $\mathbb{R}^2$? Justify your answer by examining $f(0, -t)$ as $t \rightarrow +\infty$.

4. Let

$$f(x) = \frac{1}{2}(4x_1^2 + x_2^2), \quad x^{(0)} = (1, 2)^T, \quad x^{(1)} = x^{(0)} - \alpha \nabla f(x^{(0)}).$$

Recall that f is L -smooth if $\|\nabla f(u) - \nabla f(v)\| \leq L \|u - v\|$ for all u, v .

- (a) Find $\nabla f(x)$ and show *directly from this definition* that the smallest possible smoothness constant is $L = 4$.
- (b) Use $\alpha = 1/L$ to calculate $x^{(1)}$ and $f(x^{(1)})$. Compare the actual value with the descent-lemma upper bound

$$f(x^{(0)}) - \frac{1}{2L} \|\nabla f(x^{(0)})\|^2.$$

- (c) Restart from the same $x^{(0)}$ and instead use $\alpha = 3/4$. Calculate the new point and objective value. Explain why this does not contradict the fact that $-\nabla f(x^{(0)})$ is a descent direction.